# Noise or Sensitivity? Disentangling Prompt Perturbation Effects from Intrinsic Nondeterminism in Chained LLM Agents

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**Abstract**

Prompt sensitivity—the tendency of large language models to produce different outputs under semantically equivalent prompt rewordings—is well documented for single models, but its behavior in multi-agent pipelines, where one agent's output becomes the next agent's input, has not been measured. We ask whether paraphrase-induced divergence compounds with chain depth, and introduce the control condition that turns out to be decisive: repeated runs of *unperturbed* chains at temperature 0, which isolate intrinsic sampling nondeterminism from prompt-attributable variance. Across 1,470 runs of three 4-stage pipelines (summarization, research planning, code review) with 118 validated paraphrases (120 generated; 2 degenerate duplicates of their base prompt detected and excluded, with the 15 affected runs), we find that (1) apparent divergence growth with depth is largely *not* a prompt effect: unperturbed chains amplify intrinsic nondeterminism with depth, reaching a 40% catastrophic-divergence rate at depth 3 on the most open-ended task with no perturbation at all—a rate not distinguishable from perturbed runs given our control sample size ($p = 0.31$, $n_{\text{control}} = 5$), implying that intrinsic nondeterminism accounts for at minimum the majority of the observed catastrophe rate; (2) the residual paraphrase-attributable effect is modest and does not compound—excess divergence over the noise floor stays flat or declines with depth; (3) perturbation *position* matters more than count of downstream stages: perturbing the first agent yields significantly more final-output divergence than perturbing the last (0.155 vs 0.116, Mann–Whitney $p = 2.1 \times 10^{-10}$, Cliff's $\delta = 0.246$); and (4) format-constrained final stages act as attractors that repair both noise sources—our summarization pipeline's divergence *decreases* with depth (Spearman $\rho = -0.196$) and structured-output schema violations are 0%. Our results caution that studies of prompt robustness in agent pipelines must control for nondeterminism amplification, and suggest that practitioners should invest prompt-hardening effort in early-position agents and convergent final stages. Harness, prompts, raw runs, and analysis are released for reproduction.[1]

## 1 Introduction

A growing share of LLM deployments are not single calls but *chains*: an extractor feeding a summarizer feeding an editor; an analyzer feeding a reviewer feeding a report writer. Each stage consumes the previous stage's output as input, so any instability in an early stage has, in principle, the entire remaining pipeline through which to propagate. At the same time, a well-replicated line of work shows that individual LLMs are surprisingly sensitive to semantically irrelevant prompt variation—formatting, phrasing, and paraphrase changes can swing single-model benchmark performance by

[1] https://github.com/koushik717/prompt-sensitivity-chains

double digits (Sclar et al., 2024; Zhu et al., 2023; Mizrahi et al., 2024; Sun et al., 2023; Cao et al., 2024). The natural worry, voiced but to our knowledge never measured, is multiplicative: if one agent is prompt-sensitive, is a chain of four agents four times as fragile—or worse?

This paper measures that question directly, and the measurement produces a twist. The naive experiment—paraphrase one agent's system prompt, run the chain, measure embedding divergence of the final output from a canonical-prompt run—shows divergence growing with depth on two of our three tasks, with catastrophic divergence (cosine divergence $> 0.3$) spiking from 0% at depth 1 to 20% at depth 3 in the pooled data. Reported alone, that would read as confirmation of compounding prompt sensitivity.

It would be wrong. Our design includes a control arm that most prior robustness studies omit: the *same* chains, with *canonical* prompts at every position, run repeatedly at temperature 0. These unperturbed repeats are not identical—temperature-0 decoding is not deterministic in practice, with accuracy variation of up to 15% documented across nominally deterministic runs (Atil et al., 2024; Ouyang et al., 2023; He, 2025)—and the small token-level differences they contain are amplified by exactly the same chain dynamics as paraphrase perturbations. On our most open-ended task (research planning), unperturbed depth-3 chains reach a mean divergence of 0.297 and a 40% catastrophic-divergence rate; the corresponding perturbed runs reach 0.322 and 57%—not distinguishable from the controls given our control sample size (one-sided Mann–Whitney $p = 0.31$, $n_{\text{control}} = 5$). The depth-3 "catastrophe spike" is real, but at minimum the majority of it is not a prompt-sensitivity effect: it is the chain amplifying its own sampling noise. This extends the finding that subtle nondeterminism compounds across turns in multi-turn conversation (Laban et al., 2025) to the multi-agent setting, and it means that any study attributing chained-output divergence to a manipulated variable must first subtract the divergence the pipeline generates on its own.

With the noise floor subtracted, three clean findings remain.

**Paraphrase sensitivity does not compound with depth.** Excess divergence (perturbed mean minus control mean, within task $\times$ depth cells) is modest everywhere and roughly flat or declining: $+0.079 \rightarrow +0.032$ across depths $1 \rightarrow 4$ for summarization, a steady $\approx +0.025$ for code review, and within noise for research planning. The pooled raw trend (Spearman $\rho = 0.089$) is significant but tiny, and per-task trends disagree in sign (code review $+0.34$, summarization $-0.20$)—the opposite of a universal amplification law.

**Position beats depth.** Where the paraphrase lands matters more than how many stages follow it in aggregate: perturbing the first agent produces significantly more final divergence than perturbing the last (0.155 vs 0.116; Mann–Whitney $p = 2.1 \times 10^{-10}$; Cliff's $\delta = 0.246$). Because both position groups face identical intrinsic noise, this comparison is naturally noise-robust.

**Convergent stages repair.** Stages whose task is to check, edit, or format into a fixed schema act as attractors: summarization divergence *decreases* with depth as the fact-checker and editor renormalize drafts; the code-review pipeline's JSON-constrained report writer produced zero schema violations across all perturbed depth-4 runs; and the depth-3 catastrophe rate collapses from 20% to 2% at depth 4 once the constrained final stages run. This corroborates, from an orthogonal experimental angle, the observation that deeper cascades can *reduce* factual error scores (Hallucination Cascade, 2026).

**Contributions.**

1. **To our knowledge, the first controlled separation of paraphrase-induced divergence from intrinsic sampling nondeterminism in chained LLM pipelines**, showing that what appears to be compounding prompt sensitivity is largely nondeterminism amplification—unperturbed temperature-0 chains reach catastrophic divergence rates of 40% at depth 3 on open-ended tasks.

2. **A perturbation-position effect**: early-agent paraphrases cause significantly more final-output divergence than late-agent paraphrases ($p = 2.1 \times 10^{-10}$, Cliff's $\delta = 0.246$), with practical implications for where prompt-hardening effort pays off.

3. **Evidence of convergent-stage repair**: constrained final stages attenuate both noise sources, independently corroborating cascade-reduction findings from hallucination tracking.

We release the full harness (checkpointed runner, validated paraphrase sets, versioned judge rubrics, analysis pipeline), all 1,470 raw run transcripts, and the processed data.

## 2 Related Work

**Single-model prompt sensitivity.** That LLMs are sensitive to semantically irrelevant prompt variation is by now well established. PromptBench (Zhu et al., 2023) measures robustness to adversarial prompt perturbations at character, word, sentence, and semantic levels across 8 tasks; Sclar et al. (2024) show that spurious formatting features alone—separators, capitalization, spacing—induce large performance swings; Sun et al. (2023) find that instruction-tuned models degrade on unseen paraphrases of their task instructions; and Cao et al. (2024) document gaps of up to 45 points between best- and worst-performing paraphrases of the same query, arguing for worst-prompt evaluation as a lower bound. Mizrahi et al. (2024) make the methodological case that single-prompt evaluation is unreliable, from 6.5M evaluation instances. All of this work evaluates a *single* model per call. Whether sensitivity compounds, attenuates, or transforms when a prompt-sensitive model's output becomes another prompt-sensitive model's input is the question this line of work leaves open, and the one we address.

**Error propagation in multi-agent LLM systems.** A rapidly growing 2026 literature studies how errors move through agent pipelines. Closest to us, the Hallucination Cascade study (Hallucination Cascade, 2026) tracks claim-level factual inconsistency across sequential agents and finds that deeper cascades *reduce* normalized hallucination scores ($0.422 \rightarrow 0.272$ over three agents) at a small cost in factual preservation—independently corroborating, via a different measurement (claim-level factuality vs. output-embedding divergence), the convergent-stage repair we observe. From Spark to Fire (From Spark to Fire, 2026) models collaboration as a dependency graph and shows that a single injected atomic error seed can propagate to system-level false consensus, identifying cascade amplification and topological sensitivity as vulnerability classes. MAS-FIRE (MAS-FIRE, 2026) injects 15 fault types into MAS pipelines for reliability evaluation, and AgentAsk (AgentAsk, 2025) attributes MAS underperformance to errors at inter-agent message handoffs, proposing edge-level clarification. The key difference in our design: these works perturb with *errors*—injected faults, tracked hallucinations, corrupted messages. Our perturbation is *benign by construction* (validated semantically equivalent paraphrases), so any divergence we measure is attributable to sensitivity rather than to the propagation of a defect. And none of these works measures its pipeline's unperturbed variance floor, which our results show can dominate the measured effect.

**Nondeterminism of "deterministic" LLM inference.** Atil et al. (2024) systematically document accuracy variation of up to 15% (and best-to-worst output gaps up to 70%) across runs of LLMs configured to be deterministic; Ouyang et al. (2023) report the same phenomenon for ChatGPT code generation. The mechanism is now well understood: serving-time batch-size variation interacts with non-batch-invariant kernels (He, 2025). Most relevantly, Laban et al. (2025) show that in multi-turn conversation, temperature-0 decoding still yields high unreliability because subtle nondeterminism compounds across tokens and turns. Our control condition extends this compounding result from the multi-turn single-agent setting to multi-agent chains—and quantifies it per task and depth, showing it grows to catastrophic levels (40% of unperturbed depth-3 runs on our most open-ended task) exactly where a naive reading would attribute the failures to prompt sensitivity.

**Positioning.** The sensitivity literature perturbs prompts but tests single models; the cascade literature tests chains but perturbs with errors; the nondeterminism literature perturbs nothing but stops at single calls or single-agent conversations. To our knowledge no prior work runs benign prompt perturbations through multi-stage pipelines *against a matched unperturbed control*, which is precisely the design needed to separate the two variance sources—and, in our data, the separation reverses the naive conclusion.

# 3 Method

## 3.1 Pipelines and tasks

Three 4-role pipelines, truncatable at depth $k \in \{1, 2, 3, 4\}$ (agent $k$'s output is the final output):

| Task | Chain | Final output |
|---|---|---|
| summarization | extractor → summarizer → fact_checker → editor | ∼100-word summary |
| research_plan | scoper → planner → critic → finalizer | structured plan document |
| code_review | analyzer → reviewer → security_checker → report_writer | JSON report |

**Table 1:** Task pipelines. The code-review report must be a JSON object with keys `summary`, `issues`, `severity`.

Five fixed inputs per task, held constant across all conditions.

## 3.2 Perturbations

Ten paraphrases per base system prompt (120 total) were generated by `claude-sonnet-4-6` at temperature 0.9 and validated by embedding similarity (all-MiniLM-L6-v2 cosine $\geq 0.85$ vs. base). Six paraphrases below threshold were manually reviewed and accepted (stylistic reframing, meaning preserved; review documented in the repository). A post-hoc identity check found 2 of the 120 generated "paraphrases" to be byte-identical (after whitespace/case normalization) to their base prompts; these were excluded, and the 15 runs perturbed with them were dropped from the perturbed condition (they are unperturbed chains), leaving 118 valid paraphrases and 1,335 analyzed perturbed runs. Exactly **one** agent's prompt is perturbed per run, at position first, middle ($\lfloor \text{depth}/2 \rfloor$), or last (deduplicated per depth).

### 3.3 Conditions

- **baseline** — canonical prompts at every position; the reference run per (task, input, depth).
- **perturbed** — one position paraphrased: 3 tasks $\times$ 5 inputs $\times$ 10 paraphrases $\times$ $\sum$ positions(depth) = 1,350 runs (15 excluded as pseudo-perturbed per §3.2; **1,335 analyzed**).
- **control** — canonical prompts, 5 repeats at temperature 0 per (task, depth) on one fixed input = 60 runs; quantifies intrinsic nondeterminism (the noise floor).

Total: 1,470 runs / 4,200 agent calls; zero failed runs after retries; agents `claude-haiku-4-5-20251001` at temperature 0, max 1,024 tokens; total agent-call cost $13.68 (from logged per-call token counts).

### 3.4 Metrics

- **Divergence**: $1-$ cosine similarity (all-MiniLM-L6-v2) between the final outputs of a perturbed (or control) run and its same-(task, input, depth) baseline run.
- **Catastrophic divergence**: divergence $> 0.3$ (transcript inspection confirms this threshold separates rewordings from semantic derailment; see §5).
- **Excess divergence**: perturbed cell mean $-$ control cell mean within (task, depth)—the paraphrase-attributable component.
- **Task success**: LLM-as-judge (`claude-sonnet-4-6`, temperature 0), depth-appropriate versioned rubrics (v2) grading the artifact the depth-$k$ chain actually produces; judge model $\neq$ agent model; 10% double-judged $\rightarrow$ 96.3% exact agreement, mean $|\Delta|$ 0.04 ($n = 164$).
- **Schema violations**: JSON parse + required-key check (code_review, depth 4 only).

Statistics: bootstrap 95% CIs (5,000 resamples); Spearman trend tests on divergence and on the catastrophe indicator; Mann–Whitney $U$ + Cliff's $\delta$ for the position contrast. Effect sizes are reported alongside $p$-values throughout.

## 4 Results

### 4.1 Raw divergence grows weakly—and not uniformly—with depth

Pooled across tasks, mean divergence rises from 0.0996 [95% CI 0.0908, 0.1084] at depth 1 to 0.1272 [0.1186, 0.1364] at depth 4—a 1.28× amplification, with a Spearman trend of $\rho = 0.089$ ($p = 1.2 \times 10^{-3}$): statistically detectable, but far from the multiplicative compounding the fragility hypothesis predicts (Fig. 1). The pooled number moreover conceals qualitative disagreement between tasks (Fig. 2): divergence *rises* with depth for code review ($\rho = 0.339$, $p = 1.5 \times 10^{-13}$) and research planning ($\rho = 0.225$, $p = 2.2 \times 10^{-6}$) but *falls* for summarization ($\rho = -0.196$, $p = 2.8 \times 10^{-5}$), whose fact-checking and editing stages progressively pull perturbed drafts back toward the baseline. There is no universal amplification law to report; the sign of the depth effect is a property of the pipeline's stage structure, not of chaining itself.

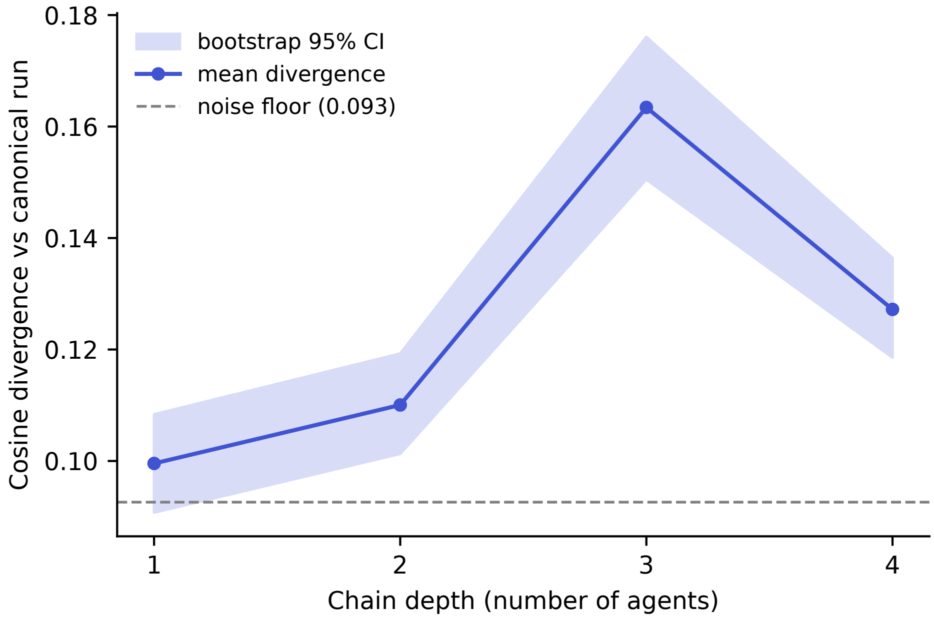


**Figure 1:** Mean embedding divergence of perturbed runs vs. chain depth, pooled across tasks, with bootstrap 95% CIs.

## 4.2 The catastrophe spike is largely nondeterminism, not sensitivity

Pooled catastrophe rate: 0% (d1) $\rightarrow$ 2.3% (d2) $\rightarrow$ 20.0% (d3) $\rightarrow$ 2.3% (d4); the depth-3 spike is driven entirely by research_plan (57.2%). Control chains at the same cell: mean divergence 0.297, catastrophe rate 40%, vs. perturbed 0.322—not distinguishable given the control sample size (one-sided Mann–Whitney $p = 0.31$, $n_{\text{control}} = 5$). At minimum, intrinsic nondeterminism accounts for the majority of the observed catastrophe rate at this cell; our control arm is too small to bound any residual paraphrase contribution tightly.

Notably, the noise floor is strongly *task-dependent*: control divergence is near zero for summarization at every depth (0.005–0.032), moderate for code review (0.072–0.126), and large and depth-growing for research planning (0.066 $\rightarrow$ 0.297 across d1$\rightarrow$d3). The same chain architecture, run on the same infrastructure at the same temperature, produces intrinsic output variance spanning nearly two orders of magnitude depending on how open-ended the stage artifacts are. This is itself a methodological result: single-number robustness claims about "LLM chains" are underdetermined without specifying the task's constraint structure, because the denominator against which any perturbation effect must be judged varies by task and depth.

## 4.3 Excess (paraphrase-attributable) divergence does not compound

Subtracting each (task, depth) cell's control mean from its perturbed mean isolates the component of divergence attributable to the paraphrase. This excess is modest everywhere and shows no depth amplification in any task: for summarization it *declines* from +0.079 (d1) to +0.032 (d4); for code review it holds nearly constant at $\approx$+0.025 across all four depths; for research planning it fluctuates within the noise (+0.038 at d1 to −0.037 at d4—a negative excess, i.e. perturbed runs diverging *less* than unperturbed controls at depth 4). With $n = 5$ controls per cell these excesses are descriptive rather than tightly estimated, but their pattern is uniform: whatever divergence paraphrases inject,

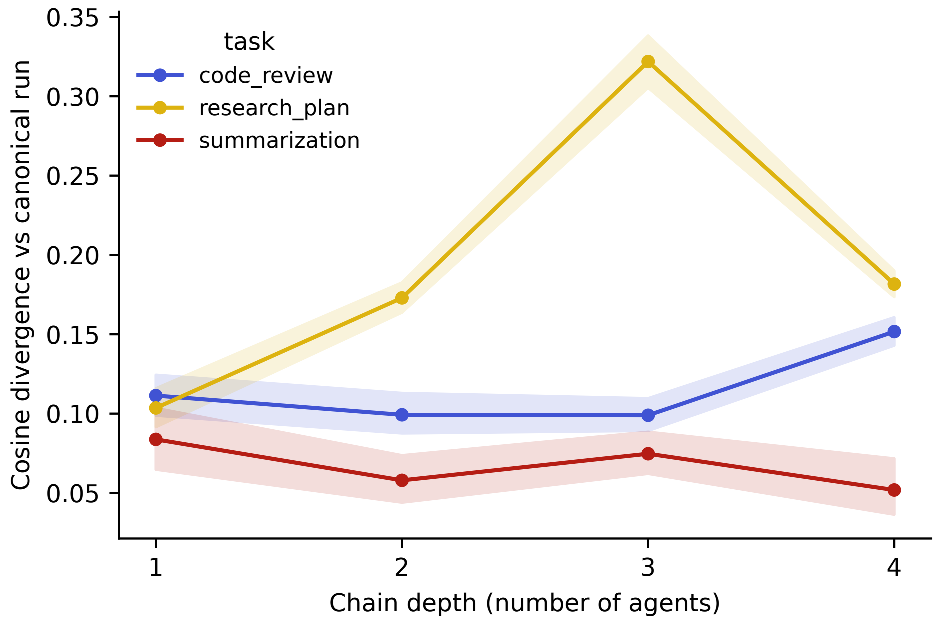


**Figure 2:** Per-task divergence vs. depth. The sign of the depth trend differs by task: rising for code review and research planning, falling for summarization.

the chain does not multiply it.

### 4.4 Position, not depth, is where prompt wording matters

Restricting to depths $\geq 2$ where both positions exist, perturbing the *first* agent yields a mean final divergence of 0.1554 ($n = 450$) against 0.1163 ($n = 440$) for the *last* agent—a significant difference (Mann–Whitney $U$, $p = 2.1 \times 10^{-10}$) with a small-to-medium effect size (Cliff's $\delta = 0.246$). Restricting to depth $\geq 3$ where all three positions exist, the ordering is monotone in position: first $0.169 >$ middle $0.138 >$ last 0.129. Because first- and last-position groups within a cell face identical intrinsic noise, this contrast is naturally robust to the noise-floor concerns of §4.2. The practical reading: a paraphrase that lands early has its perturbation propagated, reinterpreted, and built upon by every subsequent stage, while a late paraphrase can only re-style the nearly finished artifact.

### 4.5 Convergent stages repair both noise sources

Three independent observations triangulate the same mechanism. First, the pooled catastrophe rate collapses from 20.0% at depth 3 to 2.3% at depth 4—adding the final, most format-constrained stage (editor / finalizer / report writer) *reduces* catastrophic divergence. Second, summarization's negative depth trend (§4.1) shows repair acting on paraphrase perturbations specifically. Third, the structured-output check is perfect: 0% schema violations across all perturbed depth-4 code-review runs—no paraphrase anywhere in the chain ever broke the final JSON contract. Judge scores tell the same story from the quality side (Fig. 3): mean score dips at depth 3 (7.34 [7.20, 7.48]) exactly where the open-ended critic stage maximizes divergence, then recovers at depth 4 (7.85 [7.71, 7.99]). Constrained convergent stages act as attractors in output space—a cheap architectural lever for chain reliability, and one that corroborates the cascade-depth hallucination reduction reported by

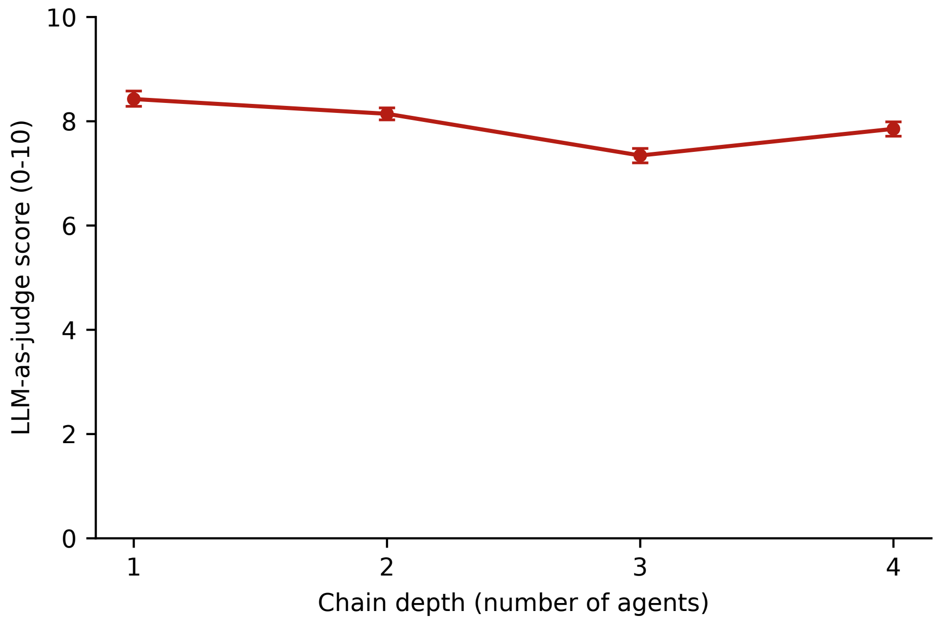


**Figure 3:** LLM-as-judge success scores (0–10, depth-appropriate rubrics) vs. depth. The dip at depth 3 coincides with the open-ended critic stage; scores recover at depth 4 when the constrained finalizer runs.

Hallucination Cascade (2026) from a different measurement angle.

## 5 Case study: a cascade failure

Run `summarization|sum_4|d4|p2|middle` (divergence 1.010, the grid maximum): a paraphrase of the *fact-checker* prompt led it to output an analysis *about* the draft rather than a corrected draft; the downstream editor, receiving analysis instead of a summary, declined to edit and asked for the missing summary—producing a final output unrelated to the source article. The failure is qualitative and structural (a role-contract violation between stages), not a gradual drift. Full transcripts of the failed and baseline runs are included in the released data.

## 6 Limitations

- Single agent-model family (Claude Haiku 4.5) and one embedding model (MiniLM); effect sizes may differ across model families and embedders.
- Three tasks, five inputs each; task heterogeneity in our own results warns against over-generalizing any single trend.
- The control arm is small (5 repeats per task × depth cell; 60 runs total)—sufficient to reveal the noise floor's magnitude and depth trend, not to tightly bound it. The perturbed-vs-control comparison at the critical cell is correspondingly low-powered ($n_{\text{control}} = 5$).
- The catastrophe threshold (0.3) is a single operating point chosen from transcript inspection; results are reproducible under nearby thresholds with the released analysis code.

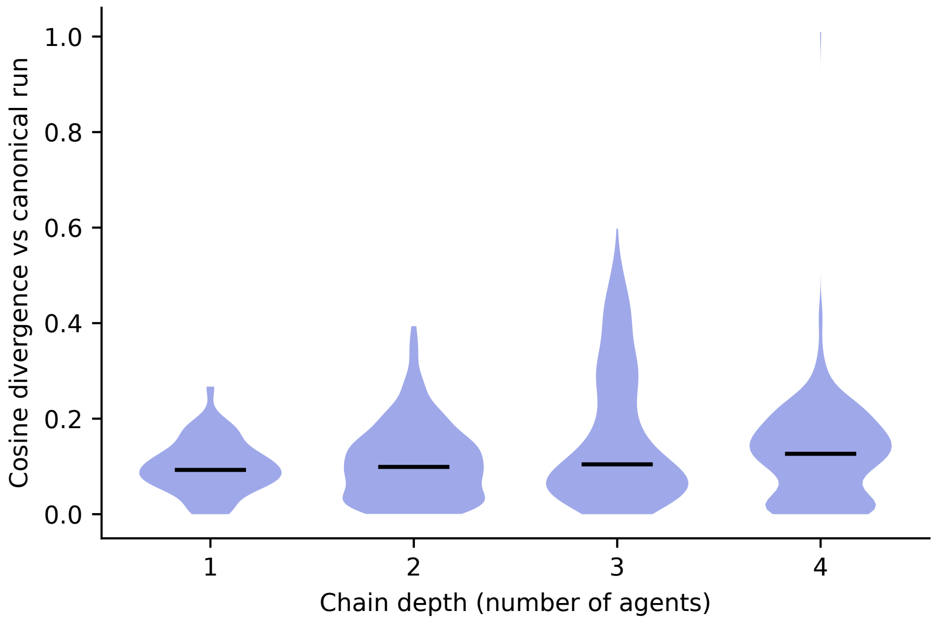


**Figure 4:** Distribution of divergence by depth (violin plots). Depth increases the tails far more than the bulk of the distribution.

- Paraphrases were LLM-generated and initially LLM-validated (a potential circularity for an LLM-sensitivity study, documented in the repository); a full human review of all 118 active paraphrases was subsequently performed by an author, confirming meaning preservation.
- LLM-as-judge scores, though reliable (96.3% exact agreement), share training lineage with the agent model's vendor.
- Our chains are limited to 4 stages; whether the position effect and convergent-stage repair we observe persist, saturate, or reverse in much longer chains (tens to hundreds of agents) is an open question we did not test. Large-scale multi-agent systems have been studied at this scale for coordination quality and emergent behavior—e.g. a 25,000-run study of coordination protocols across 4–256 agents (Anonymous, 2026d) and MegaAgent's autonomous systems scaling to hundreds of agents (Wang et al., 2024)—but, to our knowledge, no prior work measures how a benign prompt perturbation propagates through chains at this scale. Whether the position effect strengthens, saturates, or is swamped by other noise sources as chain length grows by an order of magnitude or more is a natural and, we think, important direction for future work.

# 7 Conclusion

In four-stage LLM pipelines, semantically equivalent prompt rewordings do not compound into runaway divergence: chains amplify their own sampling nondeterminism far more than they amplify paraphrase perturbations, and constrained late stages repair both. Robustness engineering for agent chains should therefore (a) control for intrinsic nondeterminism before attributing divergence to any manipulated variable, (b) invest prompt-hardening effort in early-position agents, and (c) treat constrained, convergent final stages as a cheap reliability mechanism.

## Acknowledgments

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